

SigLA: Signal-Language-Action Agents with Reinforcement Learning for Time-Series Early Anomaly Warning

中文题目： SigLA: 面向时间序列提前异常警报的信号-语言-动作强化学习智能体

Abstract

Time-series anomaly warning should not only detect abnormal windows, but also decide when and how a monitoring system should act before an event becomes irreversible. This paper proposes **SigLA**, a precursor-aware Signal-Language-Action agent for time-series anomaly warning. SigLA introduces a precursor-window-aware formulation that treats warnings as valid only when they occur within an actionable pre-onset interval. It further constructs calibrated and disentangled anomaly profiles to describe emerging risks beyond scalar anomaly scores. By jointly reasoning over signals, anomaly evidence, logs, expert rules, and historical cases, SigLA outputs executable actions such as wait, alarm, suppress, inspect, request evidence, recalibrate, and escalate.

时间序列异常预警不应只判断窗口是否异常，还应决定系统在事件不可逆之前何时、如何采取行动。本文提出 **SigLA**，一种面向时间序列异常预警的前兆感知信号-语言-动作智能体。**SigLA** 引入前兆窗口感知设定，仅将在可操作事件前窗口内触发的报警视为有效预警；同时构造校准去纠缠的异常画像，以刻画超越单一异常分数的风险证据。通过联合推理时间序列信号、异常证据、系统日志、专家规则和历史案例，**SigLA** 输出等待、报警、压制、检查、请求证据、重校准和升级等可执行监控动作。

1 Introduction

Time-series anomaly detection has been widely studied in industrial monitoring, cyber-physical systems, cloud services, spacecraft telemetry, finance, and healthcare. Most existing methods formulate the task as identifying whether an observed time point or window is anomalous. Under this setting, a model usually outputs an anomaly score or a binary normal/anomalous label. Although such outputs are useful for retrospective detection, they are often insufficient for real monitoring systems, where the central question is not only whether an anomaly exists, but also what action should be taken.

In practical monitoring scenarios, operators may need to alarm, wait, suppress a likely false alarm, inspect a specific variable, request additional evidence, recalibrate a threshold, or escalate the warning level. These actions have different costs and consequences. Raising an alarm too early may cause alert fatigue, while raising it too late may leave no time for intervention. Inspecting variables may improve confidence but consumes operational resources. Suppressing an alarm may reduce noise, but may also hide a real risk. Therefore, anomaly warning is naturally a decision-making problem rather than a pure score-thresholding problem.

A further challenge is that early warning should be evaluated by its practical intervention value. A warning raised before the event onset is not necessarily useful if it occurs only moments before

the event. Many existing evaluation protocols may still count such alarms as pre-onset detections, even though human operators or automated control systems may no longer have enough time to respond. This paper therefore adopts a precursor-aware view of anomaly warning: a valid warning should be triggered within an actionable pre-onset interval, neither too early to be unreliable nor too late to be useful.

To support such warning decisions, scalar anomaly scores are not enough. A score may indicate that the current window is abnormal, but it does not explain what kind of abnormality is emerging. A short spike, a persistent level shift, a seasonal or spectral break, a contextual deviation, and a correlation-structure change may require different monitoring actions. Moreover, raw statistical evidence can be misleading. Normal fluctuations may falsely activate anomaly concepts, and a single dominant anomaly may trigger multiple overlapping concept responses. For example, a local spike may also increase level-shift or seasonal-deviation scores, although the primary mechanism is still a spike.

This paper proposes **SigLA**, a precursor-aware Signal-Language-Action agent for time-series anomaly warning. The method transfers the vision-language-action paradigm to time-series monitoring: visual observations are replaced by multivariate signals, language inputs are replaced by logs, expert rules, alarm descriptions, variable semantics, and retrieved historical cases, and actions correspond to executable monitoring decisions. SigLA is not designed as another anomaly detector. Instead, it sits above detectors and converts signal evidence, anomaly profiles, language context, and historical experience into operational actions.

The first component of SigLA is a precursor-window-aware warning formulation. It distinguishes valid early warnings from premature alarms, late warnings, and post-onset detections. This formulation prevents the model from relying on observations that are already too close to the event onset and encourages the agent to act within a practically useful intervention window.

The second component is calibrated and disentangled anomaly profiling. Instead of directly using raw statistical indicators as final explanations, SigLA treats them as evidence features and constructs structured anomaly profiles, including spike, level shift, seasonal break, contextual deviation, and correlation break. Target-domain calibration reduces false concept activation caused by normal fluctuations, while disentanglement separates primary anomaly concepts from secondary responses and suppressed concepts. This provides a more reliable state representation for action selection.

The third component is a Signal-Language-Action policy. Given temporal signals, anomaly scores, calibrated anomaly profiles, system logs, expert rules, alarm descriptions, and retrieved historical cases, the agent outputs executable actions such as wait, alarm, suppress, inspect a variable, request more evidence, recalibrate a threshold, or escalate the warning level. The policy is initialized from supervised action traces and further optimized with cost-aware offline reinforcement learning or preference feedback. The objective balances valid early warning, false alarms, late warnings, missed events, inspection cost, alert flooding, and decision reliability.

SigLA also produces grounded rationales for its decisions. Instead of generating generic explanations, the rationale is required to refer to signal evidence, anomaly profiles, relevant

variables, time regions, logs, rules, or historical cases. This makes the output more suitable for operational monitoring, where users need to understand why a specific action is preferred over alternatives.

The main contributions are as follows:

1. **Precursor-aware anomaly warning.** This paper introduces a precursor-window-aware formulation that evaluates warnings by whether they occur within an actionable pre-onset interval, rather than merely before event onset.
2. **Calibrated anomaly profiling.** SigLA constructs structured anomaly profiles to describe emerging risks such as spike, level shift, seasonal break, contextual deviation, and correlation break, with calibration and disentanglement to reduce misleading concept activation.
3. **Signal-Language-Action decision agent.** SigLA transforms signals, anomaly evidence, logs, rules, alarm descriptions, and historical cases into executable monitoring actions instead of only producing anomaly scores or binary labels.
4. **Cost-aware policy optimization.** The action policy is trained from supervised traces and optimized with offline reinforcement learning or preference feedback to balance early warning utility, false alarms, late warnings, inspection cost, alert flooding, and decision reliability.
5. **Grounded warning rationales.** SigLA generates evidence-grounded rationales that connect actions with signal changes, anomaly profiles, operational context, and retrieved historical cases.

时间序列异常检测广泛应用于工业监控、信息物理系统、云服务、航天器遥测、金融和医疗等场景。现有研究通常将该任务定义为判断某个时间点或时间窗口是否异常。在这种设定下，模型一般输出异常分数或正常/异常二分类标签。这类输出对于事后检测是有价值的，但对于真实监控系统并不充分。真实系统关心的核心问题不只是当前是否异常，还包括接下来应该采取什么操作。

在实际监控场景中，运维人员可能需要报警、等待、压制疑似误报、检查某个变量、请求更多证据、重新校准阈值或升级预警等级。这些动作具有不同的成本和后果。过早报警可能造成报警疲劳，过晚报警可能错过干预窗口；检查变量可以提高决策置信度，但会消耗运维资源；压制报警可以降低噪声，但也可能掩盖真实风险。因此，异常预警本质上不是简单的分数阈值判断问题，而是一个面向成本和时机的动作决策问题。

进一步来看，早期预警的价值不应只由报警是否发生在事件 onset 之前来衡量。如果报警仅比事件发生早几个时间步，虽然形式上属于事件前报警，但人类运维人员或自动控制系统可能已经没有足够时间进行干预。现有评估方式可能仍会将这类报警计为提前检测，从而高估模型的实际预警价值。因此，本文采用前兆感知的异常预警视角：有效预警应发生在具有实际干预意义的事件前窗口内，既不能早到证据不足，也不能晚到失去处置价值。

为了支持这种预警决策，单一异常分数是不够的。异常分数可以说明当前窗口是否偏离正常模式，但无法说明正在出现的是哪类异常风险。短时 spike、持续性 level shift、季节性

或频谱结构破坏、上下文偏离以及变量相关结构变化，往往对应不同的监控动作。与此同时，原始统计证据本身也可能产生误导。正常波动可能错误激活某些异常概念，一个主导异常也可能诱发多个概念分数同时升高。例如，一个局部 spike 可能同时提高 level-shift 或 seasonal-deviation 分数，但其主导机制仍然是 spike。

本文提出 **SigLA**，一种面向时间序列异常预警的前兆感知信号-语言-动作智能体。该方法将视觉-语言-动作范式迁移到时间序列监控领域：视觉观测被多变量时间序列信号替代，语言输入对应系统日志、专家规则、报警描述、变量语义和检索到的历史案例，动作则对应可执行的监控决策。**SigLA** 不是另一个异常检测器，而是位于检测器之上，将信号证据、异常画像、语言上下文和历史经验转化为可执行运维动作。

SigLA 的第一个核心组成是前兆窗口感知的预警设定。该设定区分有效早期预警、过早报警、迟滞预警和事件后检测，避免模型依赖过于临近事件 onset 的观测来获得虚高的预警性能，并促使智能体在真正具有干预价值的时间窗口内采取行动。

SigLA 的第二个核心组成是校准去纠缠的异常画像。与直接把原始统计指标作为最终解释不同，**SigLA** 将统计证据、傅里叶证据、小波证据、上下文残差和相关结构偏移视为证据特征，并进一步构造结构化异常画像，包括 spike、level shift、seasonal break、contextual deviation 和 correlation break。目标域正常参考校准用于降低正常波动造成的概念误激活，去纠缠机制用于区分主导异常概念、伴随响应和被抑制的误激活概念，从而为动作选择提供更可靠的状态表示。

SigLA 的第三个核心组成是信号-语言-动作策略。给定时间序列信号、异常分数、校准后的异常画像、系统日志、专家规则、报警描述和检索到的历史案例，智能体输出等待、报警、压制、检查变量、请求更多证据、重新校准阈值或升级预警等级等可执行动作。动作策略首先通过监督动作轨迹进行初始化，再通过代价敏感的离线强化学习或偏好反馈进行优化，以平衡有效提前预警、误报、迟滞预警、漏报、检查成本、报警泛滥和决策可靠性。

此外，**SigLA** 输出基于证据的决策理由。该理由不是泛泛生成的文本解释，而需要指向具体的信号证据、异常画像、相关变量、时间区域、系统日志、专家规则或历史案例。对于真实监控系统而言，这一点尤为重要，因为用户不仅需要知道系统建议升级预警，还需要知道为什么升级优于等待、压制或请求更多证据。

本文的主要贡献如下：

1. **前兆感知的异常预警设定。** 本文提出前兆窗口感知的异常预警设定，根据报警是否发生在具有实际干预价值的事件前窗口内评价预警有效性，而不是仅判断报警是否早于事件 onset。
2. **校准异常画像构造。** **SigLA** 构造结构化异常画像，用于刻画 spike、level shift、seasonal break、contextual deviation 和 correlation break 等风险证据，并通过目标域校准和概念去纠缠降低误激活与重叠响应。

3. **信号-语言-动作决策智能体。** SigLA 将时间序列信号、异常证据、系统日志、专家规则、报警描述和历史案例转化为可执行监控动作，而不是只输出异常分数或二分类标签。
4. **代价敏感的策略优化。** SigLA 的动作策略由监督轨迹初始化，并通过离线强化学习或偏好反馈优化，在提前预警收益、误报、迟滞预警、检查成本、报警泛滥和决策可靠性之间进行权衡。
5. **基于证据的预警理由。** SigLA 生成与信号变化、异常画像、运维上下文和历史案例对齐的动作理由，使预警结果更适合真实运维场景中的理解、审核和执行。

2 Related Work

2.1 Time-Series Anomaly Detection and Early Warning

时间序列异常检测通常被定义为识别偏离正常模式的时间点或时间窗口。近年来，深度时间序列异常检测方法主要包括预测误差、重构误差、注意力差异、对比表征、图结构建模和通用预训练等路线。Anomaly Transformer 利用关联差异刻画异常点与正常点在时序关联上的差异 (Xu et al., 2022)；TranAD 使用 Transformer 和自条件机制进行多变量异常检测与诊断 (Tuli et al., 2022)；DCdetector 通过双注意力对比学习增强异常表征区分能力 (Yang et al., 2023)；DADA 进一步探索跨数据集的通用时间序列异常检测器，通过多域预训练提高 zero-shot 异常检测能力 (Shentu et al., 2025)。相关综述也指出，现有 TSAD 方法已经在模型结构、学习范式和基准数据方面形成较完整体系，但主流任务仍然以 anomaly score 或 normal/anomaly label 为核心输出 (Wang et al., 2025; Darban et al., 2024)。

然而，真实监控系统中的需求并不止于识别异常片段。对于工业系统、云服务和信息物理系统，异常检测结果需要进一步转化为报警、等待、检查、压制或升级等操作。早期异常预测和前兆检测开始关注事件发生前的风险信号。例如，Precursor-of-Anomaly detection 将任务从“当前是否异常”扩展为“当前是否出现未来异常的前兆” (Jhin et al., 2023)；IGCL 研究无监督异常预测，通过生成异常前兆模式来区分正常片段与可能演化为异常的前兆片段 (Zhao et al., 2025)；TFAP 则利用时间-频率对比学习和上下文建模捕捉弱前兆动态 (Li et al., 2026)。这些工作推动了从事后检测到事件前预测的转变，但大多仍然输出风险分数或预测标签，缺少面向监控系统的可执行动作空间。

SigLA 与上述方法不同。它不是提出另一个底层 anomaly detector，而是在异常检测器和运维响应之间引入一个决策智能体。SigLA 接收信号、异常分数、异常画像和语言上下文，并输出 wait、alarm、suppress、inspect、request evidence、recalibrate 和 escalate 等动作。因此，SigLA 将时间序列异常预警从“分数检测”进一步推进到“动作决策”。

2.2 Precursor Modeling and Event-Level Evaluation

早期预警的关键不只是提前发现异常，而是在具有实际干预价值的时间窗口内做出预警。传统点级指标或简单的 point-adjusted F1 容易高估异常检测性能。已有研究指出，时间序列异常检测的评价协议可能存在偏差，尤其是 point adjustment 可能使模型在异常区间内只命中一个点也获得较高分数，从而无法真实反映检测时机和检测质量（Kim et al., 2022）。也有研究系统梳理了时间序列异常检测评价指标，强调不同指标对点级、区间级和事件级检测的敏感性不同（Sørbo and Ruocco, 2024）。Benchmark 研究也表明，统一实验环境、数据集构造和评价指标对于公平比较异常检测方法非常关键（Paparrizos et al., 2022; Paparrizos et al., 2025）。

对于异常预警而言，事件级评价比点级评价更重要。模型是否在 event onset 之前报警、报警提前量是否足够、是否过早扰动、是否迟滞报警，都直接影响实际运维价值。现有 precursor detection 和 anomaly prediction 工作已经开始关注事件前信号，但通常仍把“是否早于 onset”作为主要判断，而较少显式区分有效可干预窗口、过早报警和迟滞预警（Jhin et al., 2023; Zhao et al., 2025; Li et al., 2026）。

SigLA 因此引入 precursor-window-aware early warning formulation。该设定将事件前时间区域进一步划分为稳定正常区、有效前兆窗口和迟滞区。只有在有效前兆窗口内触发的报警才被视为有效早期预警；过早报警会造成报警疲劳，过晚报警则失去干预价值。该设定使 SigLA 的优化目标不再是单纯降低 detection delay，而是学习何时采取具有实际处置意义的动作。

2.3 Anomaly Profiling and Explainable Time-Series Evidence

从异常检测走向动作决策，需要回答一个更细的问题：当前异常风险是什么类型。单一 anomaly score 无法区分短时 spike、持续性 level shift、seasonal or spectral break、contextual deviation 或 correlation break，而这些异常形态往往对应不同运维动作。已有时间序列异常检测工作中，模型通常通过重构误差、预测误差或注意力差异产生异常分数，但这些分数本身并不直接给出可操作的异常画像（Xu et al., 2022; Tuli et al., 2022; Yang et al., 2023）。

可解释异常检测和 LLM 辅助异常检测开始关注异常解释。AnomalyLLM 使用大语言模型指导知识蒸馏，使学生模型学习 LLM 教师网络的特征表达，并通过教师-学生差异检测异常（Liu et al., 2024）。LLMAD 进一步尝试直接利用 LLM 进行 in-context time-series anomaly detection，并通过检索相似时间序列片段和 anomaly detection chain-of-thought 提供可解释结果（Liu et al., 2025）。Zhou and Yu 系统分析了 LLM 是否能够理解时间序列异常，指出 LLM 对简单异常具有一定识别能力，但对复杂真实场景的细粒度异常理解仍然有限（Zhou and Yu, 2025）。

这些研究说明语言模型和解释机制对 TSAD 有潜力，但 SigLA 的异常画像并不依赖 LLM 直接“看图说异常”。SigLA 构造 calibrated and disentangled anomaly profile，将统计证据、傅里叶证据、小波证据、上下文残差和变量关系偏移转化为结构化异常概念。其目的不是

单纯生成解释文本，而是为动作策略提供可靠状态表示。例如，短时 spike 更可能触发 wait、suppress 或 inspect；持续 level shift 更可能触发 alarm 或 escalate；correlation break 则更适合检查相关变量。这样，异常画像成为连接检测证据和动作决策的中间层。

2.4 Signal-Language and Multimodal Time-Series Reasoning

近年来，时间序列分析开始与语言模型、多模态模型和检索机制结合。时间序列本身通常缺少语义上下文，但真实监控系统中往往同时存在系统日志、专家规则、报警说明、变量描述、维护记录和历史案例。LLMAD 表明，检索相似的正负时间序列片段可以提升 LLM 在异常检测中的解释性和判断能力（Liu et al., 2025）。AnomalyLLM 说明大语言模型可以作为知识教师，为少样本时间序列异常检测提供表示监督（Liu et al., 2024）。Zhou and Yu 进一步指出，LLM 对时间序列异常的理解方式与输入表达形式密切相关，图像化、文本化和数值化输入会影响模型表现（Zhou and Yu, 2025）。

与这些工作相比，SigLA 的目标不是把时间序列转换成文本后交给 LLM 分类，也不是只用语言模型生成解释。SigLA 将语言视为 operational context：系统日志说明当前工况，专家规则提供安全约束，报警描述解释变量含义，历史案例提供相似事件经验。这些语言信息与信号表征和异常画像共同构成 agent state。换言之，SigLA 的 signal-language fusion 服务于动作选择，而不仅是服务于 anomaly classification。

这种设计也不同于普通 multimodal time-series reasoning。普通多模态方法通常关注融合信号、文本或图像以提升预测准确率，而 SigLA 进一步要求融合结果能够支持可执行动作。它需要回答的不只是“当前是否异常”，还包括“是否应该报警”“应该检查哪个变量”“是否需要更多证据”“是否应该升级预警”。因此，SigLA 更接近面向监控系统的 signal-language-action reasoning。

2.5 Agentic Decision-Making and Cost-Aware Policy Optimization

视觉-语言-动作模型和语言智能体研究表明，感知、语言理解和动作生成可以在统一框架中建模。SayCan 将语言模型的任务规划能力与机器人技能可行性结合，使输出动作既符合指令又能在环境中执行（Ahn et al., 2023）。PaLM-E 将视觉和连续状态输入接入语言模型，用于具身推理和机器人任务（Driess et al., 2023）。RT-2 进一步提出 vision-language-action 模型，将机器人动作表示为 token，并通过视觉-语言预训练增强机器人策略泛化能力（Zitkovich et al., 2023）。ReAct 则展示了语言模型可以交替生成 reasoning traces 和 actions，通过推理与外部行动交互改进任务求解过程（Yao et al., 2023）。

SigLA 借鉴这一范式，但将其从机器人控制迁移到时间序列监控。这里的 perception 不是图像，而是多变量信号和异常画像；language 不是普通 caption，而是日志、规则、报警描述和历史案例；action 不是机械臂控制，而是报警、等待、压制、检查、请求证据、重校准和升级等监控动作。因此，SigLA 是 Signal-Language-Action agent，而不是 Vision-Language-Action agent 的简单类比。

由于监控动作具有成本和后果，SigLA 的动作策略需要进行代价敏感优化。Offline reinforcement learning 为不能在真实系统中自由探索的场景提供了可行训练范式。CQL 通过保守 Q 函数降低离线 RL 中 out-of-distribution action 的过度估计风险（Kumar et al., 2020）；IQL 避免显式查询数据集外动作价值，使离线策略学习更稳定（Kostrikov et al., 2022）。对于包含语言理由或偏好反馈的动作策略，DPO 提供了一种不显式训练 reward model 的偏好优化方式（Rafailov et al., 2023）。SigLA 使用监督动作轨迹初始化策略，再通过离线强化学习或偏好反馈优化，使 agent 能在有效预警、误报、迟滞报警、检查成本、报警泛滥和解释忠实性之间进行权衡。

3 Problem Formulation

3.1 Time-Series Anomaly Warning Setting

考虑一个多变量时间序列流 $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T\}$ ，其中每个观测 $\mathbf{x}_t \in \mathbb{R}^d$ ， d 表示变量数量。给定窗口长度 w ，监控系统在时间 t 只能观察当前及历史窗口 $\mathbf{X}_{t-w+1:t}$ ，不能使用未来观测：

$$\mathbf{X}_{t-w+1:t} = [\mathbf{x}_{t-w+1}, \mathbf{x}_{t-w+2}, \dots, \mathbf{x}_t].$$

异常事件集合记为 $\mathcal{E} = \{e_i\}_{i=1}^N$ ，每个事件 e_i 具有事件起点 T_i 。事件起点只在训练标签构造、离线回放和评价阶段使用；在测试推理时，SigLA 只能基于当前可见的信号、分数、日志、规则、历史案例和历史动作状态做出决策。与普通 TSAD 不同，本文关注的不是单个窗口是否异常，而是监控系统在风险演化过程中应该采取什么动作。底层检测器可以输出异常分数，但 SigLA 的目标是把检测证据转化为可执行的预警动作。

3.2 Precursor-Window-Aware Early Warning

对于事件 e_i ，设 T_i 为事件起点， $L_{\max}$ 为最大有效提前量， $L_{\min}$ 为最小干预提前量。 $L_{\max}$ 用于避免过早报警， $L_{\min}$ 用于保证报警之后仍有足够干预时间。由此，事件前后区域被划分为稳定正常区、有效前兆窗口、迟滞预警区和事件后区域：

$$\mathcal{S}_i = \{t \mid t < T_i - L_{\max}\},$$

$$\mathcal{P}_i = \{t \mid T_i - L_{\max} \leq t \leq T_i - L_{\min}\},$$

$$\mathcal{D}_i = \{t \mid T_i - L_{\min} < t < T_i\},$$

$$\mathcal{A}_i^{\text{post}} = \{t \mid t \geq T_i\}.$$

其中 $\mathcal{P}_i$ 是有效前兆窗口。只有在 $\mathcal{P}_i$ 内触发的报警才被视为有效早期预警。若报警发生在 $\mathcal{S}_i$ 中，则可能属于过早报警；若报警发生在 $\mathcal{D}_i$ 中，即使它仍早于事件起点，也被视为迟滞预警，因为实际干预时间已经不足；若报警发生在 $\mathcal{A}_i^{\text{post}}$ 中，则属于事件后检测。该设定避免模型依赖临近事件起点的明显异常信号获得虚高预警性能。

令 τ_i 表示事件 e_i 的首次报警时间。有效预警指示函数定义为：

$$I_{\text{valid}}(\tau_i) = \mathbb{I}[T_i - L_{\max} \leq \tau_i \leq T_i - L_{\min}].$$

若事件前没有有效前兆窗口内报警，则该事件被视为 missed early warning。该定义把“早于事件起点”进一步细化为“早于事件起点且处于可操作干预窗口内”。

3.3 Observation, State, and Action Space

在每个时间 t ，SigLA 接收多源观测，包括当前信号窗口、底层检测器分数轨迹、异常画像、语言上下文、检索到的历史案例以及过去动作历史：

$$o_t = (\mathbf{X}_{t-w+1:t}, \mathbf{s}_{t-w+1:t}, \mathbf{c}_t, \mathbf{L}_t, \mathbf{R}_t, \mathbf{h}_t).$$

其中 $\mathbf{s}_{t-w+1:t}$ 表示一个或多个底层检测器产生的 anomaly score trajectory。如果共有 M 个检测器，则可写为：

$$\mathbf{s}_{t-w+1:t} = [s_{t-w+1:t}^{(1)}, s_{t-w+1:t}^{(2)}, \dots, s_{t-w+1:t}^{(M)}].$$

$\mathbf{c}_t$ 表示校准去纠缠后的 anomaly profile； $\mathbf{L}_t$ 表示语言上下文，包括系统日志、专家规则、报警描述、变量语义和运行上下文； $\mathbf{R}_t$ 表示检索到的历史相似案例； $\mathbf{h}_t$ 表示过去动作和状态摘要。这样，SigLA 的 Language 部分由 $\mathbf{L}_t$ 和 $\mathbf{R}_t$ 共同构成，而不是被删除。

SigLA 将观测映射为决策状态：

$$\mathbf{z}_t = f_{\theta}^{\text{state}}(o_t).$$

动作空间包含可执行监控动作，而不是异常分数阈值：

$$\mathcal{U} = \{\text{wait}, \text{alarm}, \text{suppress}, \text{inspect}(v), \text{request_evidence}, \text{recalibrate}(m), \text{escalate}(k)\}.$$

其中 $\text{inspect}(v)$ 表示检查变量 v ， $\text{recalibrate}(m)$ 表示重新校准第 m 个检测器或阈值， $\text{escalate}(k)$ 表示将预警升级到等级 k 。动作可以采用分层设计：第一层选择动作类型，第二层选择动作参数。

3.4 Objective

SigLA 的目标是在离线回放环境或偏好反馈环境中学习动作策略，使其在有效提前预警、误报、迟滞预警、漏报、检查成本、报警泛滥和理由忠实性之间取得平衡。策略定义为：

$$\pi_{\theta}(u_t | \mathbf{z}_t), \quad u_t \in \mathcal{U}.$$

总体目标为最大化折扣累计回报：

$$\max_{\pi_{\theta}} \mathbb{E}_{\pi_{\theta}} \left[\sum_{t=1}^T \gamma^t r_t \right].$$

其中 reward 由事件级预警效用和动作级成本共同构成：

$$r_t = \lambda_1 U_{\text{valid}} - \lambda_2 C_{\text{premature}} - \lambda_3 C_{\text{late}} - \lambda_4 C_{\text{miss}} - \lambda_5 C_{\text{inspect}} - \lambda_6 C_{\text{flood}} + \lambda_7 U_{\text{faithful}}.$$

其中 U_{valid} 奖励在有效前兆窗口内报警, $C_{\text{premature}}$ 惩罚过早报警, C_{late} 惩罚迟滞预警, C_{miss} 惩罚漏报, C_{inspect} 表示检查或请求证据的成本, C_{flood} 表示重复报警和报警疲劳成本, U_{faithful} 奖励与证据一致的理由。

4 Methodology: SigLA

4.1 Framework Overview

SigLA 包含五个主要阶段: Perceive、Profile、Read、Decide 和 Explain。Perceive 阶段编码多变量信号和底层异常分数; Profile 阶段构造校准去纠缠的异常画像; Read 阶段编码语言上下文 $\mathbf{L}_t$ 和检索历史案例 $\mathbf{R}_t$; Decide 阶段融合信号、分数、画像、语言和历史状态并输出动作; Explain 阶段生成基于证据的动作理由。

与传统 TSAD 模型不同, SigLA 不直接替代底层检测器。它位于检测器之上, 将检测证据和运维上下文转化为动作策略。该结构使底层检测能力、前兆窗口约束、异常画像和动作优化可以分别建模, 同时在最终策略中统一决策。

4.2 Signal and Score Encoder

给定窗口 $\mathbf{X}_{t-w+1:t}$ 和检测器分数轨迹 $\mathbf{s}_{t-w+1:t}$, 信号编码器提取局部动态、变量间关系和分数变化趋势。编码器可以采用 TCN、Transformer、PatchTST-style encoder 或其他时间序列表征模型。输入不仅包含原始信号, 也包含 score trend 和 uncertainty features:

$$\mathbf{h}_t^{\text{sig}} = f_{\text{sig}}(\mathbf{X}_{t-w+1:t}, \mathbf{s}_{t-w+1:t}, \Delta \mathbf{s}_{t-w+1:t}, \mathbf{u}_t^{\text{unc}}).$$

其中 $\Delta \mathbf{s}_{t-w+1:t}$ 表示分数趋势, $\mathbf{u}_t^{\text{unc}}$ 表示不确定性或多检测器分歧。该表示回答的是“当前信号是否正在偏离正常状态”, 但它本身还不能说明异常类型和动作选择。

4.3 Calibrated and Disentangled Anomaly Profiling

为了把异常检测证据转化为可操作状态, SigLA 构造异常画像 $\mathbf{c}_t$ 。该模块首先从窗口中提取可解释 evidence features, 包括 Hampel 幅值、一阶差分、峰值突出度、小波局部高频能量、傅里叶频谱距离、主频偏移、周期能量损失、上下文残差和相关矩阵偏移:

$$\mathbf{e}_t = [e_t^{\text{amp}}, e_t^{\text{diff}}, e_t^{\text{peak}}, e_t^{\text{wave}}, e_t^{\text{freq}}, e_t^{\text{context}}, e_t^{\text{corr}}].$$

这些证据特征不是最终异常概念, 而是轻量 concept extractor 的输入。extractor 输出五类 coarse anomaly concept 的 raw profile, 包括 spike、level shift、seasonal break、contextual deviation 和 correlation break:

$$\mathbf{p}_t = g_{\phi}(\mathbf{e}_t), \quad \mathbf{p}_t \in [0,1]^5.$$

由于不同系统的正常波动强度不同, raw profile 需要在目标域正常参考窗口上校准。对于每个 concept k , 在正常参考窗口中计算中位数 μ_k^{norm} 和 MAD σ_k^{norm} , 并得到校准分数:

$$\tilde{p}_t^k = \frac{p_t^k - \mu_k^{norm}}{\sigma_k^{norm} + \epsilon}.$$

校准解决的是正常波动造成的 false concept activation。例如，正常周期峰值或工况切换可能提高 spike 或 seasonal 相关分数，但若这种变化属于目标域正常参考范围，则校准后不应被视为异常概念。

进一步地，一个真实异常可能同时触发多个 concept 分数。为区分主导异常机制和伴随响应，SigLA 为每个 concept 计算 candidate dominance score:

$$d_t^k = \text{Normalize}(\alpha_k \tilde{p}_t^k + \beta_k v_t^k),$$

其中 v_t^k 表示与 concept k 对应的验证证据。对于 spike，需要局部高幅值、高导数、小波局部性强、持续时间短且快速恢复；对于 level shift，需要左右窗口中位数差异、CUSUM、持续性和变化后稳定性；对于 seasonal break，需要傅里叶频谱距离、主频偏移、周期能量损失和频谱熵；对于 correlation break，需要控制单变量 outlier 后仍存在的变量关系变化。

为抑制重叠响应，SigLA 对候选主导 concept 执行 intervention 或 adjustment，并重新计算其他 concept 的分数：

$$\begin{aligned} \mathbf{X}_t^{(k)} &= \mathcal{I}_k(\mathbf{X}_{t-w+1:t}), \\ \Delta_t^{k \rightarrow j} &= p_t^j - \text{Score}_j(\mathbf{X}_t^{(k)}). \end{aligned}$$

如果 $\Delta_t^{k \rightarrow j}$ 较大，则说明 concept j 的升高可能是 concept k 的伴随响应，而不是独立异常机制。最终异常画像包含 calibrated profile、primary concept、secondary responses、co-existing concepts、suppressed concepts 和 supporting evidence:

$$\mathbf{c}_t = (\tilde{\mathbf{p}}_t, k_t^{primary}, \mathcal{C}_t^{sec}, \mathcal{C}_t^{co}, \mathcal{C}_t^{sup}, \mathbf{e}_t).$$

4.4 Precursor-Aware Risk Accumulation

早期预警通常依赖连续弱信号，而不是单个时间点的强异常。因此，SigLA 在异常画像之上构造风险累积状态 ρ_t ，用于描述系统是否正在从稳定正常状态进入前兆风险阶段：

$$\rho_t = \alpha \rho_{t-1} + (1 - \alpha) q(\mathbf{h}_t^{sig}, \mathbf{c}_t, \mathbf{s}_{t-w+1:t}).$$

其中 $q(\cdot)$ 表示由信号表示、异常画像和分数轨迹计算得到的局部风险， α 控制历史风险的衰减。与普通 anomaly score 不同， ρ_t 强调风险是否持续演化，并为动作策略提供“等待、请求证据、报警或升级”的时机依据。

4.5 Language and Historical Memory Encoder

语言上下文 $\mathbf{L}_t$ 包括系统日志、专家规则、报警描述、变量语义、维护记录和当前工况说明。历史记忆 $\mathbf{R}_t$ 包括检索到的相似正常片段、相似误报片段和相似真实事件片段。SigLA 使用 frozen language encoder、轻量 instruction model 或检索增强编码器表示这些信息：

$$\mathbf{h}_t^{lang} = f_{lang}(\mathbf{L}_t), \quad \mathbf{h}_t^{ret} = f_{ret}(\mathbf{R}_t).$$

语言与历史记忆的作用不是替代信号判断，而是补充 operational context。例如，某个变量升高在特定工况下可能正常，但在另一个日志事件之后可能表示风险；某类历史案例也可以提示当前模式更像误报还是早期故障。

4.6 Signal-Language-Profile Fusion

SigLA 将信号表示、检测器分数、异常画像、风险累积状态、语言上下文和历史记忆融合为最终决策状态。融合可以采用 gated fusion、cross-attention 或 instruction-conditioned attention：

$$\mathbf{z}_t = f_{fuse}(\mathbf{h}_t^{sig}, \mathbf{s}_{t-w+1:t}, \mathbf{c}_t, \rho_t, \mathbf{h}_t^{lang}, \mathbf{h}_t^{ret}, \mathbf{h}_t).$$

这种融合使动作策略能够同时利用数值证据、异常类型、风险演化、规则约束和历史经验。

4.7 Hierarchical Action Policy

动作策略采用分层输出。第一层选择动作类型，第二层选择动作参数。例如，当动作类型为 inspect 时，动作参数是变量 v ；当动作类型为 recalibrate 时，动作参数是待校准的 detector 或 threshold m ；当动作类型为 escalate 时，动作参数是预警等级 k ：

$$\pi_{\theta}(u_t | \mathbf{z}_t) = \pi_{\theta}^{type}(a_t | \mathbf{z}_t) \cdot \pi_{\theta}^{arg}(b_t | a_t, \mathbf{z}_t).$$

其中 a_t 表示动作类型， b_t 表示动作参数。分层策略可以降低动作空间复杂度，并使动作理由更容易与异常画像对齐。

4.8 Cost-Aware Policy Optimization

SigLA 的训练包括三个阶段。第一阶段是 supervised action trace learning，用专家规则、历史处置记录、oracle simulator 或事件级 oracle 生成动作轨迹，并用行为克隆初始化策略：

$$\mathcal{L}_{BC} = - \sum_t \log \pi_{\theta}(u_t^E | \mathbf{z}_t).$$

第二阶段是 cost-aware offline reinforcement learning。离线环境由历史序列、事件起点、动作成本和前兆窗口规则构成，策略在回放数据上优化长期 reward，而不是在真实系统中无约束在线试错：

$$\mathcal{J}_{RL}(\theta) = \mathbb{E}_{\pi_{\theta}} \left[\sum_t \gamma^t r_t \right].$$

第三阶段是 **preference optimization**。给定同一事件上的两个动作序列，如果一个序列更早进入有效前兆窗口、误报更少、检查成本更低且理由更可信，则它被标为 **preferred sequence**。可以使用偏好损失优化策略：

$$\begin{aligned} \Delta S_{\theta} &= S_{\theta}(\mathbf{a}_{high}) - S_{\theta}(\mathbf{a}_{low}), \\ \mathcal{L}_{pref} &= -\log \sigma(\beta \Delta S_{\theta}). \end{aligned}$$

其中 $\mathbf{a}_{high}$ 和 $\mathbf{a}_{low}$ 分别表示偏好较高和较低的动作序列， $S_{\theta}(\cdot)$ 表示策略对动作序列的打分。

4.9 Grounded Rationale Generation

SigLA 输出动作的同时生成结构化理由。理由必须引用具体证据，包括异常画像、变量、时间区域、日志、规则或历史案例。为了避免无依据解释，理由生成受到 **evidence grounding** 约束：

$$y_t = f_{exp}(u_t, \mathbf{z}_t, \mathbf{c}_t, \mathbf{L}_t, \mathbf{R}_t).$$

一个有效理由应说明当前为何选择该动作，而不是其他可选动作。例如，若 **primary concept** 为 **level shift**，风险持续累积且日志提示关键设备切换，则 **escalate** 比 **suppress** 更合理；若 **spike** 是短时、局部且快速恢复，则 **wait** 或 **inspect** 可能优于直接升级。

5 Algorithm

Algorithm 1: SigLA Training and Inference

Input: signal streams $\mathbf{X}$, detector scores $\mathbf{s}$, language contexts $\mathbf{L}$, retrieved cases $\mathbf{R}$, event onsets $\{T_i\}$, action traces $\mathcal{D}_{act}$.

Output: precursor-aware Signal-Language-Action policy π_{θ} .

Step 1. Construct stable, precursor, late-warning, and post-onset regions for each event according to T_i , L_{max} , and L_{min} .

Step 2. Train or load bottom anomaly detectors and obtain anomaly score trajectories $\mathbf{s}_{t-w+1:t}$.

Step 3. Extract interpretable evidence features $\mathbf{e}_t$ from each signal window.

Step 4. Train the lightweight concept extractor on synthetic or weakly supervised evidence-concept labels.

Step 5. Calibrate concept profiles with target-domain normal reference windows.

Step 6. Apply concept disentanglement to obtain $\mathbf{c}_t$, including primary concepts, secondary responses, co-existing concepts, and suppressed concepts.

Step 7. Encode signals, score trajectories, anomaly profiles, language contexts, and retrieved historical cases.

Step 8. Fuse all sources into the decision state $\mathbf{z}_t$.

Step 9. Initialize the action policy π_θ with supervised action traces using behavior cloning.

Step 10. Optimize the policy with cost-aware offline reinforcement learning under the precursor-window reward.

Step 11. Optionally refine the policy using preference feedback over action sequences and rationales.

Step 12. During inference, observe only current and historical information, output action u_t , and generate grounded rationale y_t .

6 Experiments

6.1 Datasets and Language Augmentation

实验可使用 SWaT、WADI、SMD、PSM、MSL 和 SMAP 等多变量时间序列异常检测基准。对于具有攻击或故障区间的数据集，事件 onset 可由异常区间起点或攻击日志近似得到。对于缺少语言上下文的数据集，可构造增强文本，包括变量含义、设备说明、攻击描述、运行阶段、专家规则模板、报警描述和检索到的历史相似案例。

需要注意，并非所有异常都具有可观测前兆。对于瞬时攻击或完全随机故障，应单独标记为 non-precursor events，避免把无法提前预警的事件误用于前兆窗口评价。

6.2 Baselines

对比方法应覆盖四类。第一类是传统 score-based policies，包括固定阈值、动态阈值、point-adjusted detector 和 event-level thresholding。第二类是 early warning baselines，包括只基于 anomaly score 的提前报警策略、risk accumulation policy 和 lead-time-aware warning policy。第三类是 agent baselines，包括 signal-only agent、language-only agent、signal-language classifier without action policy 和 flat action policy。第四类是 SigLA 的消融版本，包括去除前兆窗口、去除异常画像、去除语言上下文、去除检索历史案例和去除策略优化。

6.3 Metrics

除 AUROC、AUPRC 和 F1 外，实验应重点评价事件级和动作级指标。令 N 为事件总数， N_{valid} 为首次报警落在有效前兆窗口内的事件数， N_{late} 为首次报警落在迟滞区的事件数。Valid Early-Warning Recall 定义为：

$$\text{VEW} - \text{Recall} = \frac{N_{\text{valid}}}{N}.$$

Late Warning Rate 统计在最小干预提前量之后才报警的事件比例，用于判断模型是否依赖临近 onset 的明显异常信号：

$$\text{LWR} = \frac{N_{\text{late}}}{N}.$$

Premature Alarm Rate 统计过早报警比例，Inspection Cost 统计检查和请求证据的动作成本，Action Utility 统计包含有效预警收益、误报惩罚、迟滞惩罚和检查成本的总体动作效用。Rationale Faithfulness 评价动作理由是否与异常画像、变量证据、日志和历史案例一致。

6.4 Ablation Study

消融实验应围绕 SigLA 的三个核心创新展开。第一，去除 precursor-window-aware formulation，观察模型是否更倾向于在临近 onset 时报警，并导致 late-warning rate 上升。第二，去除 calibrated and disentangled anomaly profile，只使用 anomaly score 和语言上下文，观察动作效用和解释忠实性是否下降。第三，去除 Signal-Language-Action policy，将输出退化为 score thresholding 或 action classification，观察其在检查成本、压制误报和升级决策上的表现。

其他消融包括 w/o language context、w/o historical retrieval、w/o offline RL、w/o preference optimization、w/o inspect action、w/o recalibrate action 和 flat action policy。

6.5 Case Study

案例分析应展示一个事件从稳定正常到前兆窗口再到 onset 的全过程。理想案例包括：传统阈值策略在临近 onset 才报警，虽然命中但属于 late warning；SigLA 在有效前兆窗口内先 request evidence 或 inspect 关键变量，随后在风险持续累积时 alarm 或 escalate。案例中应同时展示异常画像变化、风险累积曲线、动作序列和 grounded rationale。

7 Conclusion

本文提出 SigLA，一种面向时间序列异常预警的前兆感知信号-语言-动作智能体。SigLA 不把异常检测结果停留在 anomaly score 或 binary label，而是进一步回答监控系统在风险演化过程中应该采取什么动作。通过 precursor-window-aware early warning formulation，SigLA 将有效预警限定在具有实际干预价值的事件前窗口内，避免模型依赖临近 onset 的明显异常信号获得虚高性能。通过 calibrated and disentangled anomaly profiling，SigLA 将原始检测证据转化为可操作的异常画像，从而支撑报警、等待、压制、检查、请求证据、重校准和升级等动作选择。通过 signal-language-action policy optimization，SigLA 将信号证据、语言上下文、历史案例和动作成本统一到一个可解释的监控决策框架中。未来

工作可以进一步研究真实工业系统中的人机协同反馈、跨域动作策略迁移和安全约束下的在线部署。

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